

Machine Learning: Introduction

ECMM422

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Outline

- 1 Machine learning tasks: Classification
- 2 Regression
- 3 Machines for learning
- 4 Learning and generalisation
- 5 Unsupervised learning
- 6 Machine Learning Paradigms

Classification

Train a machine $\mathcal{F}$ to map features to targets

$$\mathcal{F} : \mathbf{x} \mapsto t$$

$$t = \mathcal{F}(\mathbf{x}; \mathbf{w})$$

Features: $\mathbf{x} \in \mathbb{R}^n$

Inputs: $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_N$

Targets (correct outputs):

$$t_1, t_2, t_3, \dots, t_N$$

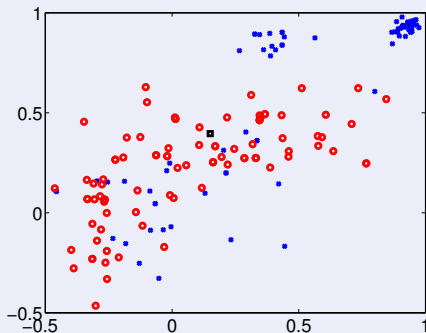
t_n is the class to which $\mathbf{x}_n$ belongs

w: Adjustable parameters controlling the behaviour of $\mathcal{F}$

Training pairs: $\{(\mathbf{x}_n, t_n)\}_{n=1}^N$

Prediction: Given new $\mathbf{x}$, the predicted class is $\hat{t} = \mathcal{F}(\mathbf{x}; \mathbf{w}^*)$

Tremor data

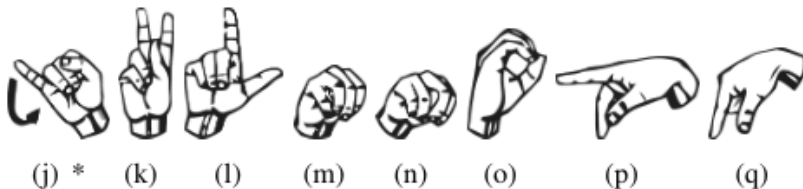
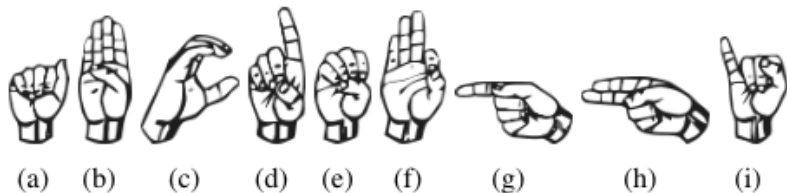


Red: Patients

Blue: Controls

About 85% accuracy possible

American Sign Language: Spelling



Aim

To develop a system that given a camera, will identify the hand shapes done by the user and the corresponding letter in the American Sign Language alphabet, and provide an interactive interface to allow spelling on a computer.

ASL Fingerspelling recognition

- ▶ Locating where the hand is located: hands are highly articulated and move fast!
- ▶ recognise the same hand shape across different subjects and lighting condition
- ▶ the difference between the signs corresponding to different letters is sometimes subtle (eg, a, e, and n).
- ▶ do this in real time.

Learning by Example

How?

We collect a large corpus of examples of multiple subjects signing each letter, then extract some features to describe their appearance and learn a separation function using decision trees to detect which sign is being made.

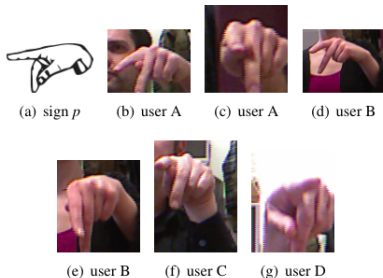


ASL: Ambiguity

Why is it hard?

There is a large level of ambiguity between the hand shapes we are trying to recognise. Some letters look the same; conversely people's hand physiology and dexterity vary, as does viewing angle, illumination, etc.

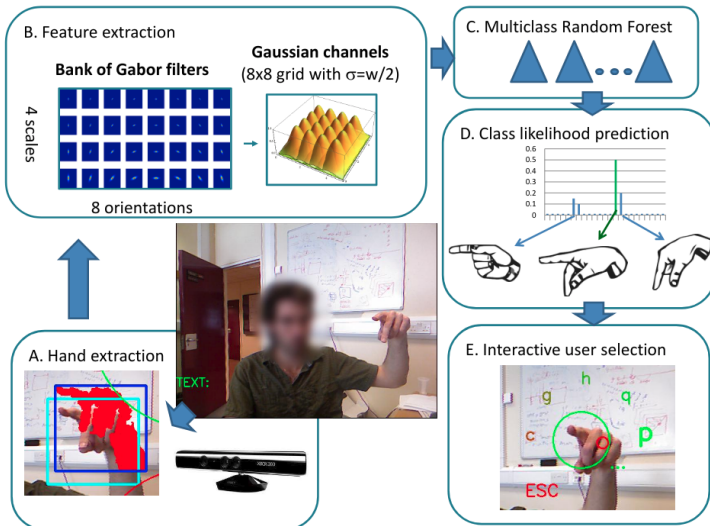
Within class



Between classes



ASL Fingerspelling: System



<https://www.youtube.com/watch?v=bgA2YVclb-c>

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Regression

Train a machine $\mathcal{F}$ to map features to continuous targets

$$\mathcal{F} : \mathbf{x} \mapsto t$$

$$t = \mathcal{F}(\mathbf{x}; \mathbf{w})$$

w: Vector of adjustable parameters that control the behaviour of $\mathcal{F}$

Inputs:

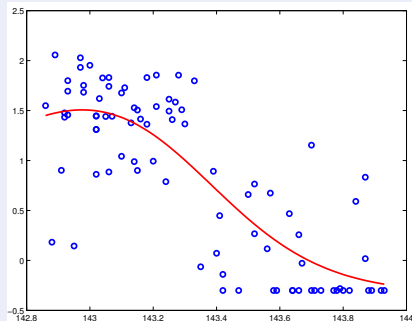
$x_1, x_2, x_3, \dots, x_N$

Targets (correct outputs):

$t_1, t_2, t_3, \dots, t_N$

Training pairs: $\{(\mathbf{x}_n, t_n)\}_{n=1}^N$

Great Barrier Reef fauna

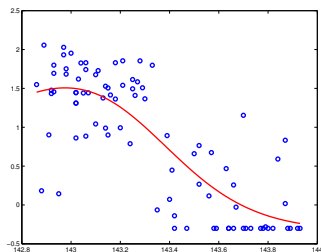
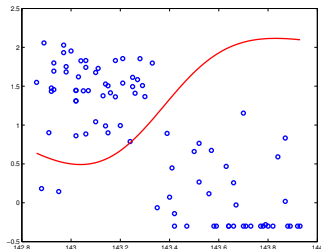
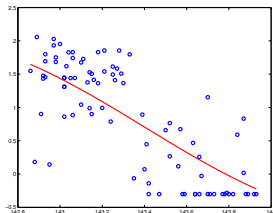


Red line marks trained neural network prediction.

Function or program: $\hat{t} = \mathcal{F}(x; w)$

- x Inputs: vector of features
- $\hat{t}$ Output: estimate of the target
- w Vector of adjustable parameters that affect the behaviour of the machine

Adjust weights to obtain best match to the data.



Training and prediction

Training data: $\mathcal{D} = \{(x_n, t_n)\}, n = 1, \dots, N$.

Adjust parameters w to minimise error function $E(w)$ between output of the machine $\mathcal{F}(x; w)$ and targets. For example, in regression problems:

$$E(w) = \frac{1}{N} \sum_{n=1}^N [t_n - \mathcal{F}(x_n; w)]^2$$

If $E(w) = 0$ then the machine gives the correct output for every datum in the training data.

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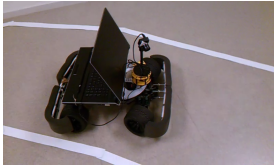
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Prediction

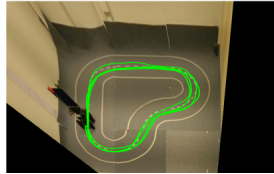
If $\mathbf{w}^*$ are the parameters that minimise E , then predict the target corresponding to unknown features $\mathbf{x}$ as

$$\hat{t} = \mathcal{F}(\mathbf{x}; \mathbf{w}^*)$$

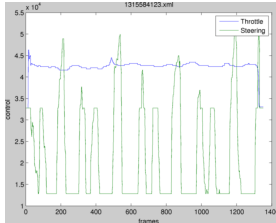
Our dataset



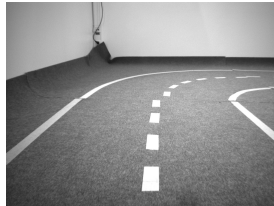
(a) the car



(b) the track



(c) recorded control



(d) camera view

Typical computer vision pipeline

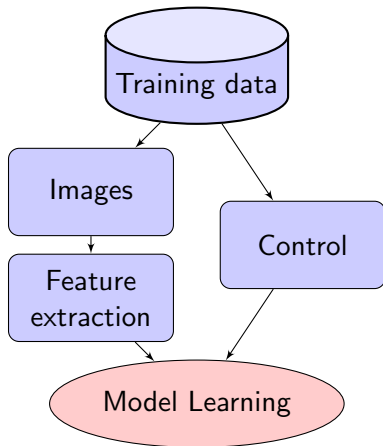


Figure: Learning stage

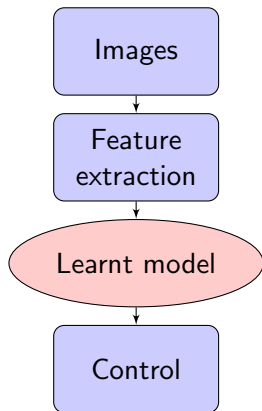
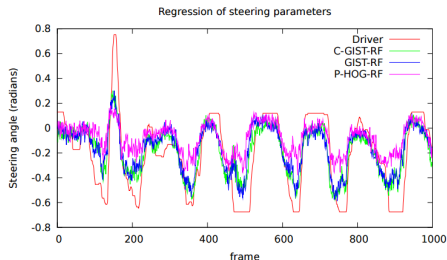
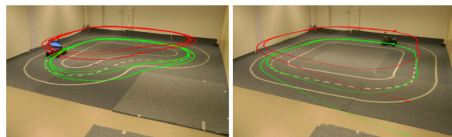


Figure: Prediction stage

Predicting steering from image features

- ▶ The system can learn an approximation of the mapping between what the driver see and how he steers.
- ▶ performance can be improved with better image features!
- ▶ One problem with RF: they tend to underestimate extrema (due to the averaging steps).



Pugeault, N. and Bowden, R. "Driving me Around the Bend: Learning to Drive from Visual Gist", In *Proc. of the IEEE Workshop on Challenges and Opportunities in Robotic Perception*, at ICCV'2011.

Outline

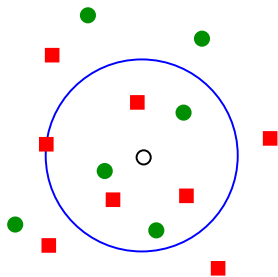
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k nearest neighbours classifier

Given a training data set of instances and targets, $\{x_n, t_n\}$, $n = 1 \dots N$.

Predict the class of a new data point x by

- 1 Find the k nearest neighbours to x in the training data.
- 2 Assign the x to the class that has the greater number of the k nearest neighbours.
(Ties are broken randomly.)



k nearest neighbours classifier

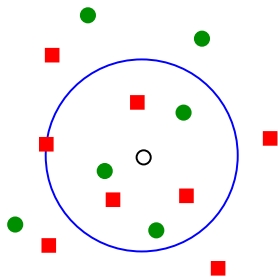
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Probability: If there are k_1 samples of class 1 among the k neighbours, the probability that x is in class 1 is estimated as

$$p = \frac{k_1}{k}$$



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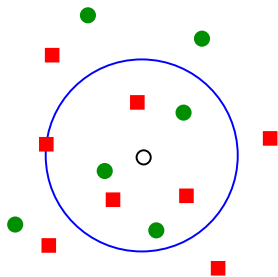
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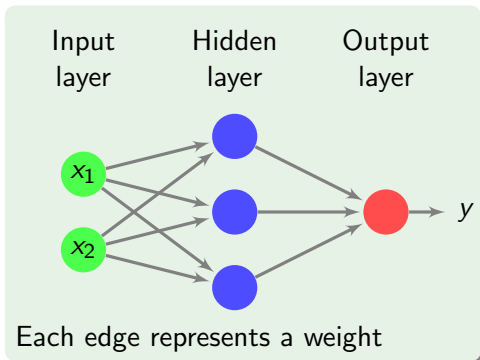
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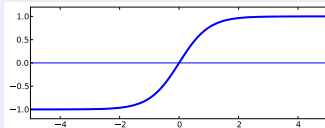
What are the parameters here? We have to know the position of all the training instances!



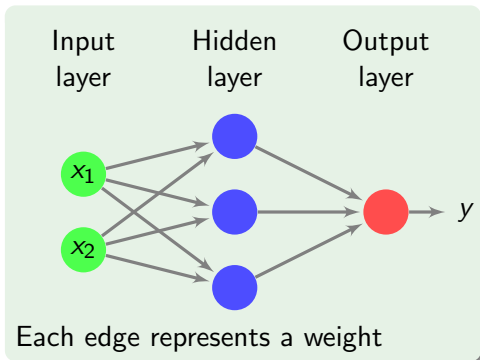
Neural networks: flexible learning machines



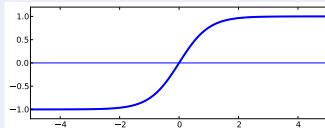
Each hidden layer node sums its weighted input and passes it through a *sigmoidal nonlinearity*



Neural networks: flexible learning machines



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The set of all weights are the parameters w that can be adjusted to affect the behaviour of the network

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Supervised learning paradigm

- ▶ Collect/observe training data $\mathcal{D} = \{(\mathbf{x}_n, t_n)\}_{n=1}^N$
- ▶ Learn parameters of a machine $\mathcal{F}(\mathbf{x}, \mathbf{w})$ by minimising an error function
- ▶ Given new instances $\hat{\mathbf{x}}$, predict the target as $\mathcal{F}(\hat{\mathbf{x}}, \mathbf{w}^*)$

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Goals of learning $\mathcal{F}(x, w)$:

- ▶ Learn the characteristics of a dataset
- ▶ Ignore the noise in the dataset

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- ▶ Learn the characteristics of a dataset
- ▶ Ignore the noise in the dataset

As number of tunable parameters in $\mathcal{F}$ becomes large:

- ▶ $\mathcal{F}$ fits *training* dataset very well
- ▶ but generalises poorly; new data is poorly fitted.

Learning with neural networks

True mapping

$$\sin(2\pi x/5) - x/20$$

Noise

Gaussian; standard deviation 0.25

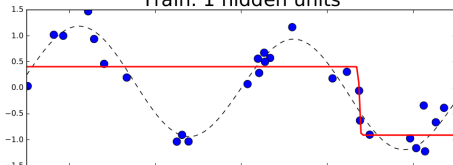
Trained

with 15 examples

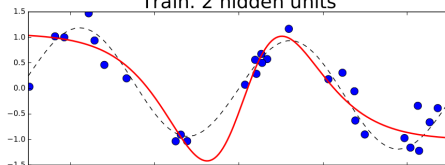
Test

with different examples

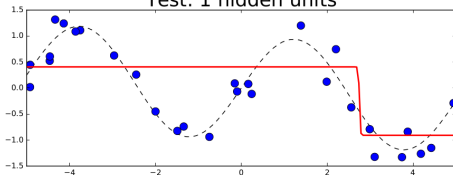
Train. 1 hidden units



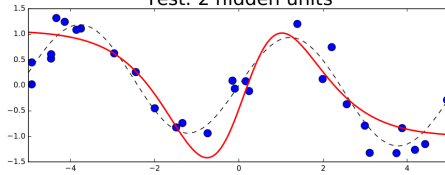
Train. 2 hidden units



Test. 1 hidden units

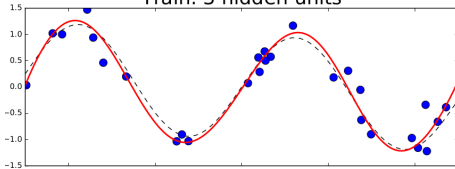


Test. 2 hidden units

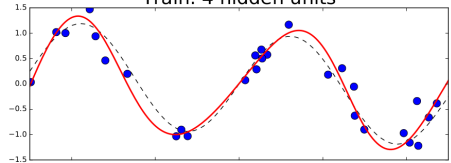


Learning with neural networks

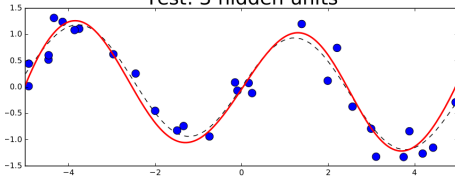
Train. 3 hidden units



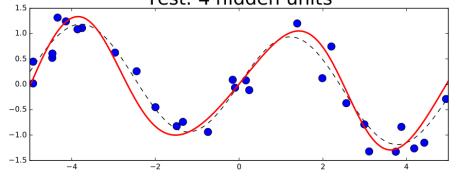
Train. 4 hidden units



Test. 3 hidden units

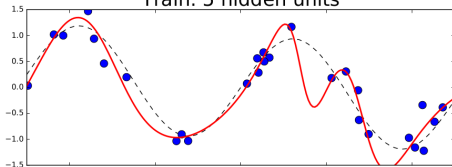


Test. 4 hidden units

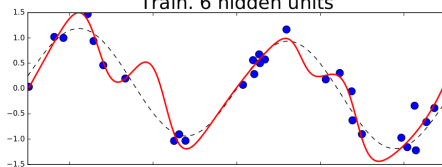


Learning with neural networks

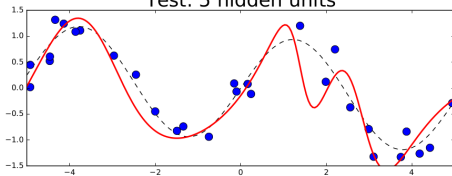
Train. 5 hidden units



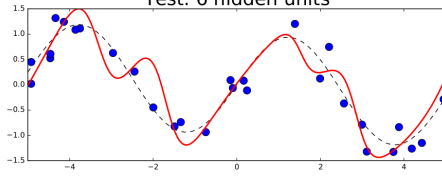
Train. 6 hidden units



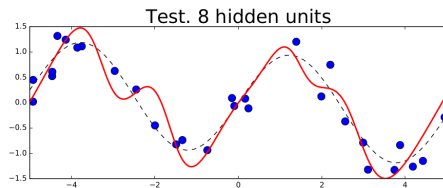
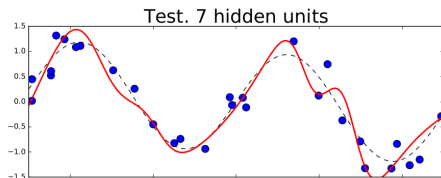
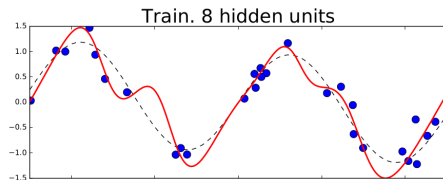
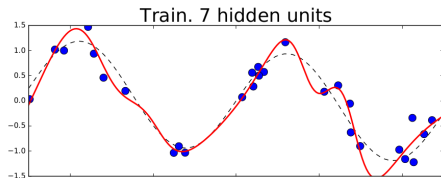
Test. 5 hidden units



Test. 6 hidden units

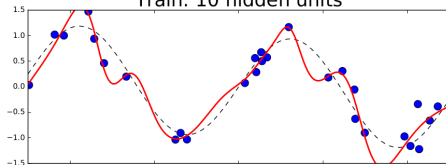


Learning with neural networks

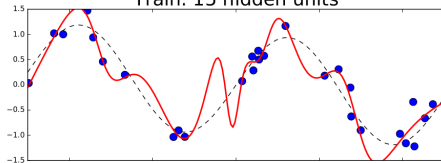


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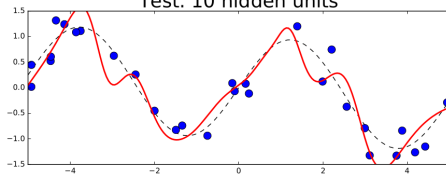
Train. 10 hidden units



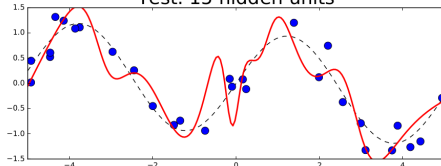
Train. 15 hidden units



Test. 10 hidden units

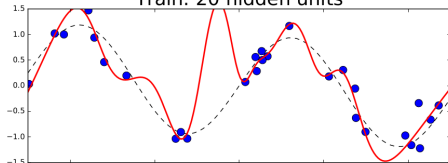


Test. 15 hidden units

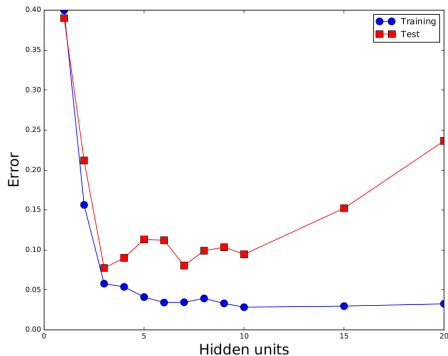
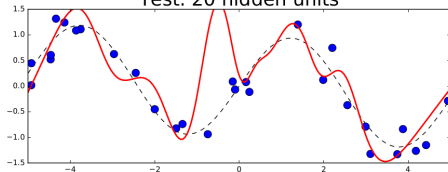


Learning with neural networks

Train. 20 hidden units



Test. 20 hidden units



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Unsupervised learning

No targets

Goal is to learn salient features of data

Clustering

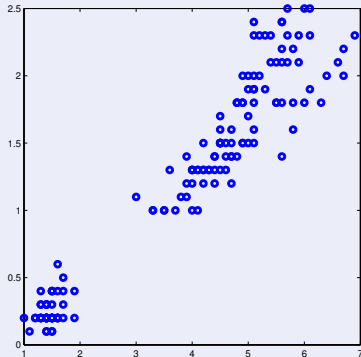
Principal components analysis

Density estimation

Blind source separation

Dimensionality reduction (MDS, SOM)

Fisher iris data



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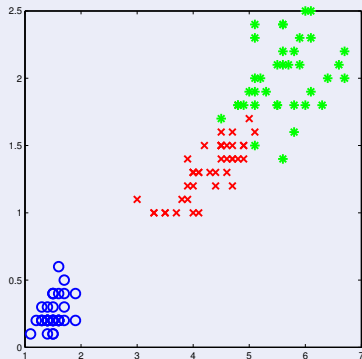
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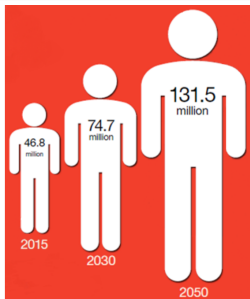
Dimensionality reduction (MDS, SOM)

Fisher iris data



Dementia

- ▶ A significant decline in one or more of the cognitive domains
- ▶ Cognitive deficits that result in a loss of independence in activities of daily living



Clustering

- ▶ Cluster dementia sufferers to understand:
 - ▶ Dementia diagnoses: robust diagnostic categories, mixed dementia types
 - ▶ Significant features
- ▶ 32,611 records; 286 variables

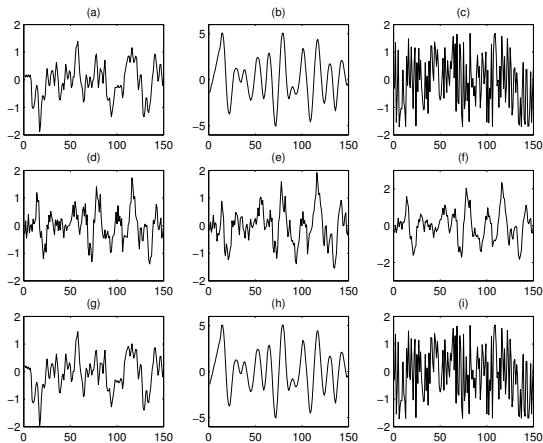
Generative Adversarial Networks

- ▶ Can we learn what the input looks like from the data?
- ▶ Learning is achieved by training two networks in competition: a *generator* and a *discriminator*.
- ▶ Learnt from CelebA face dataset: over 200,000 images of 10,000 celebrities.



Blind source separation

- ▶ How do we distinguish the speakers at a cocktail party?
- ▶ How can a machine unravel individual voices from recordings of a mixture?
- ▶ Possible by assuming that the speakers are statistically *independent*.

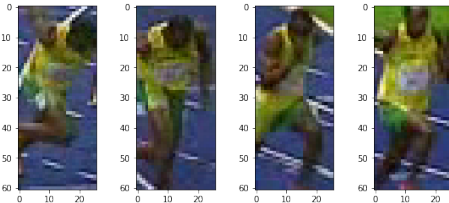


Video tracking objects that change shape and appearance



Video tracking objects that change shape and appearance

- ▶ Only thing known:
position at the start
- ▶ Target changes
appearance and moves at
the same time: template
needs to be updated.



- ▶ Problem: template drift!

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Machine learning paradigms

A machine $\mathcal{F}$ receives in input a finite sample of *instances*, $x_1, x_2, x_3, \dots, x_n$

Semi-supervised learning

The machine is given targets corresponding to a *few* instances but has access to a large amount of *unlabelled* instances.

Labelled data is often very expensive to acquire, requiring human intervention, but automatically-collected data is often cheap.

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Reinforcement learning

The machine gains *rewards* in response to *actions* it takes, but is not directly supervised by being given the “correct” result. Rewards may be accrued long after the action is taken.

Stick balancing, lift control, backgammon, landing space craft

Machine learning



**"The machine learning algorithm
wants to know if we'd like a
dozen wireless mice to feed the
Python book we just bought."**